LLMs in the Study of Human Language Processing

Dr. Fritz Günther & Dr. Francesca Capuano

TeaP 2026 · Pre-conference Workshop

March 15, 2026

University of Tübingen

Outline

Why LLMs to study human language processing

What can we extract from LLMs?

Types of questions LLMs let us ask

LLMs as research tools

Why LLMs to Study Human Language Processing

- Accessible computational systems
- Text-trained predictive learners
- Unprecedented linguistic competence

Outline

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Types of questions LLMs let us ask

LLMs as research tools

I. Prompting

- Choices
- Ratings
- Continuations

II. Probabilistic Expectations

- Next-word probability
- Surprisal
- Entropy / Uncertainty

III. Internal Representations

- Hidden states
- Contextual embeddings
- Attention patterns

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Behavioural Alignment with Humans

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- Structural priming
- Sound symbolism
- Semantic illusion

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Goal

Test whether a model reproduces **behavioral effects** observed in psycholinguistic experiments.

Behavioural Alignment with Humans

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- Sound symbolism
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Test whether a model reproduces **behavioral effects** observed in psycholinguistic experiments.

Behavioural Alignment with Humans

Structural priming

Do large language models resemble humans in language use?

Zhenguang G. Cai[1,2], Xufeng Duan[1], David A. Haslett[1], Shuqi Wang[1], Martin J. Pickering[3]
1 Department of Linguistics and Modern Languages, The Chinese University of Hong Kong
2 Brain and Mind Institute, The Chinese University of Hong Kong
3 Department of Psychology, University of Edinburgh
zhenguangcai@cuhk.edu.hk, xufeng.duan@link.cuhk.edu.hk, david.a.haslett@link.cuhk.edu.hk, ShuqiWang@link.cuhk.edu.hk, martin.pickering@ed.ac.uk

Behavioural Alignment with Humans

Structural Priming

Humans tend to reuse recently encountered syntactic structures.

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Prime sentence

The racing driver showed ***the mechanic the problem****.*

(Double-object structure)

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Target prompt

The patient showed ...

Human tendency

The patient showed ***the nurse his hand****.*

Linking / Correlational Inference

Does an LLM-derived measure predict human behavior?

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Examples

- Surprisal $\rightarrow$ Reading times/N400
- Embedding similarity $\rightarrow$ N400
- Entropy $\rightarrow$ Eye movements

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Interpretation
LLM measures may operationalize latent processing variables such as *predictability*, *compatibility*, and *uncertainty*.

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Linking / Correlational Inference

Surprisal/Embedding similarity → N400

SPECIAL ISSUE:
Cognitive Computational Neuroscience of Language

Strong Prediction: Language Model Surprisal Explains Multiple N400 Effects

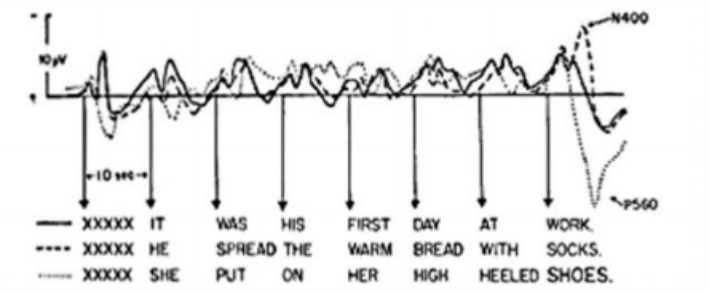
James A. Michaelov[1], Megan D. Bardolph[1], Cyma K. Van Petten[2], Benjamin K. Bergen[1], and Seana Coulson[1]

[1]Department of Cognitive Science, University of California, San Diego, La Jolla, CA, USA
[2]Department of Psychology, Binghamton University, State University of New York, Binghamton, NY, USA

Keywords: distributional semantics, ERPs, N400, neural language models, predictive coding

Linking / Correlational Inference

Surprisal/Embedding similarity → N400



Kutas & Hillyard (1980)

Predictability Account

Sentence Context

↓

Expectation about upcoming word

↓

Prediction Error

↓

N400 Amplitude

LLM Surprisal

(contextual probability)

Similarity Account

Sentence Context

↓

Activated Semantic Features

↓

Compatibility with incoming word

↓

N400 Amplitude

Embedding Cosine Similarity

(context-word compatibility)

Sufficiency / Learnability Inference

Can this phenomenon emerge from statistical text-only learning?

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- Syntactic island constraints
- Pragmatic implicature
- Agreement phenomena

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If a text-trained model captures it, strong modularity claims are **in-principle constrained**.

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If a text-trained model captures it, strong modularity claims are **in-principle constrained**.

Syntactic island constraints

Using Computational Models to Test Syntactic Learnability

Ethan Gotlieb Wilcox
Richard Futrell
Roger Levy

Sufficiency / Learnability Inference

Syntactic island constraints

Long-distance dependencies

I know ***what*** *the lion devoured __ yesterday.*

Sufficiency / Learnability Inference

Syntactic island constraints

Long-distance dependencies

*I know **what** the lion devoured __ yesterday.*

Filled-gap effect

*I know what the lion devoured **the gazelle** yesterday.*

Unexpected object → increased processing difficulty.

Sufficiency / Learnability Inference

Syntactic island constraints

Long-distance dependencies

*I know **what** the lion devoured __ yesterday.*

Filled-gap effect

*I know what the lion devoured **the gazelle** yesterday.*

Unexpected object → increased processing difficulty.

Island structures block the dependency

I know what the actress bought the painting
that depicted __ yesterday.

Model Comparison as Theory Testing

Which architecture better aligns with humans?

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- RLHF vs. base models
- Memory-augmented vs. standard

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Architectural components that improve alignment become candidate **explanatory variables**.

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Model Comparison as Theory Testing

Multimodal vs. text-only

Do Multimodal Large Language Models and Humans Ground Language Similarly?

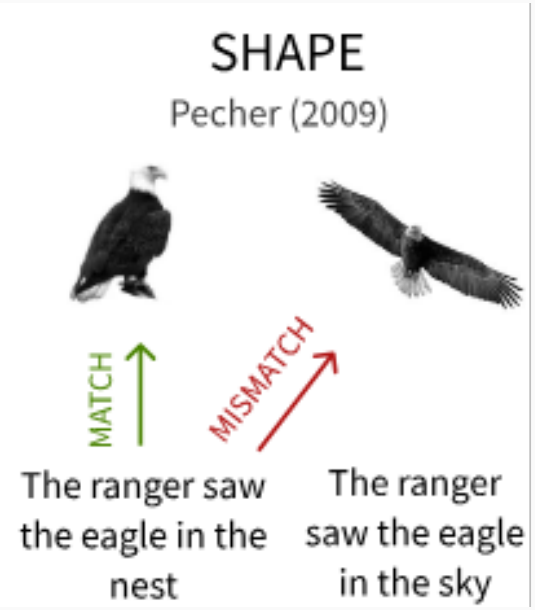
Cameron R. Jones*
University of California, San Diego
Department of Cognitive Science
cameron@ucsd.edu

Benjamin Bergen
University of California, San Diego
Department of Cognitive Science
bkbergen@ucsd.edu

Sean Trott
University of California, San Diego
Department of Cognitive Science
sttrott@ucsd.edu

Model Comparison as Theory Testing

Multimodal vs. text-only



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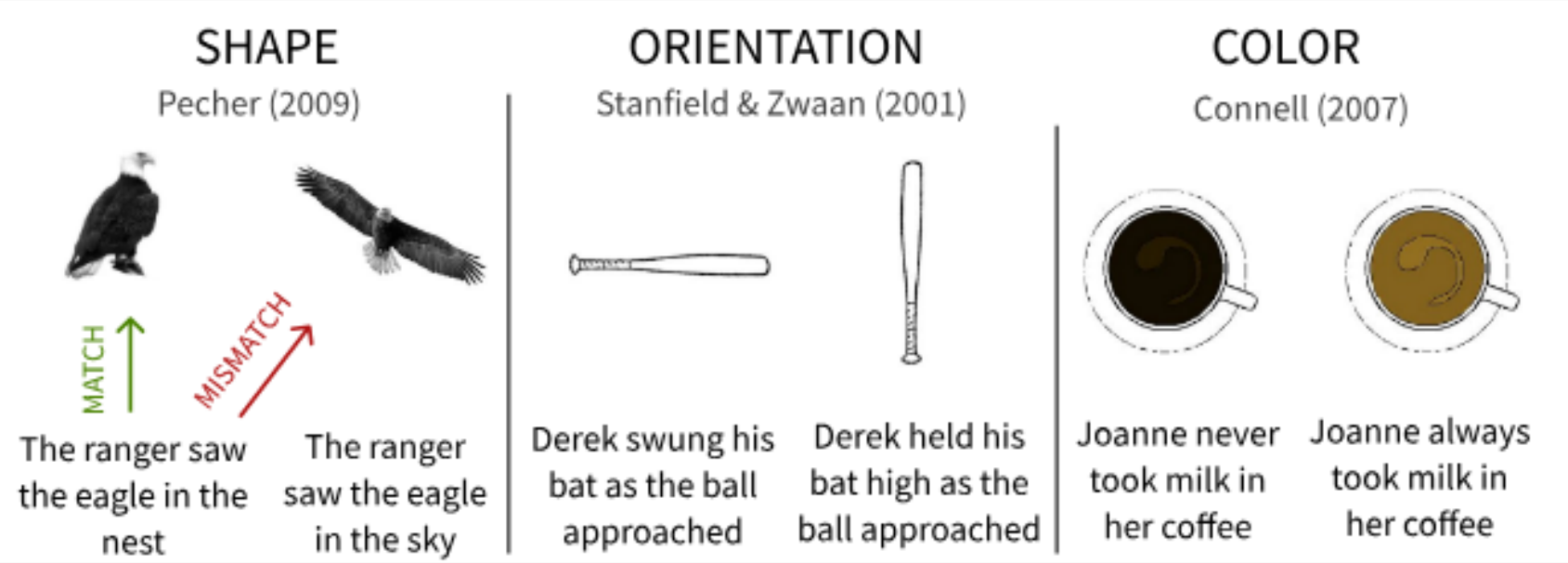

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Architectural components that improve alignment become candidate **explanatory variables**.

Model Comparison as Theory Testing

RLHF vs. base models

Pragmatic Reasoning in GPT Models: Replication of a Subtle Negation Effect

Francesca Capuano (francesca.capuano@uni-tuebingen.de)
Department of Psychology, University of Tübingen
Schleichstraße 4, 72076 Tübingen, Germany

Barbara Kaup (barbara.kaup@uni-tuebingen.de)
Department of Psychology, University of Tübingen
Schleichstraße 4, 72076 Tübingen, Germany

Model Comparison as Theory Testing

RLHF vs. base models

	GPT-3	InstructGPT	GPT-3.5 (Chat)	GPT-4 (Chat)	Humans
Experiment 3 (this)	neg >aff**	neg >aff**	n.s.	n.s.	n.s.
Experiment 6 (see)	n.s.	neg <aff*	neg >aff**	neg > aff*	neg >aff***
Experiment 7 (want)	n.s.	n.s.	n.s.	neg > aff***	neg >aff***

Representation Questions

What kind of structure is encoded internally?

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- Probing classifiers
- Distributed vs. localized encoding

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These studies test whether abstract linguistic structure is **recoverable from internal representations**, informing debates about distributed vs. symbolic encoding.

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Representation Questions

Syntactic tree decodability

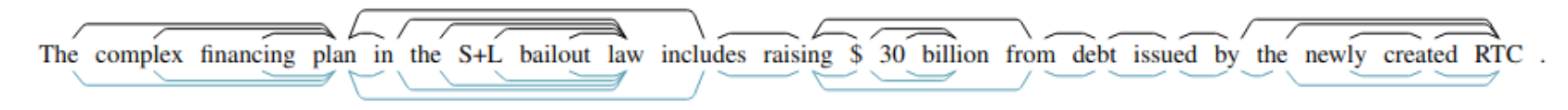
A Structural Probe for Finding Syntax in Word Representations

John Hewitt
Stanford University
johnhew@stanford.edu

Christopher D. Manning
Stanford University
manning@stanford.edu

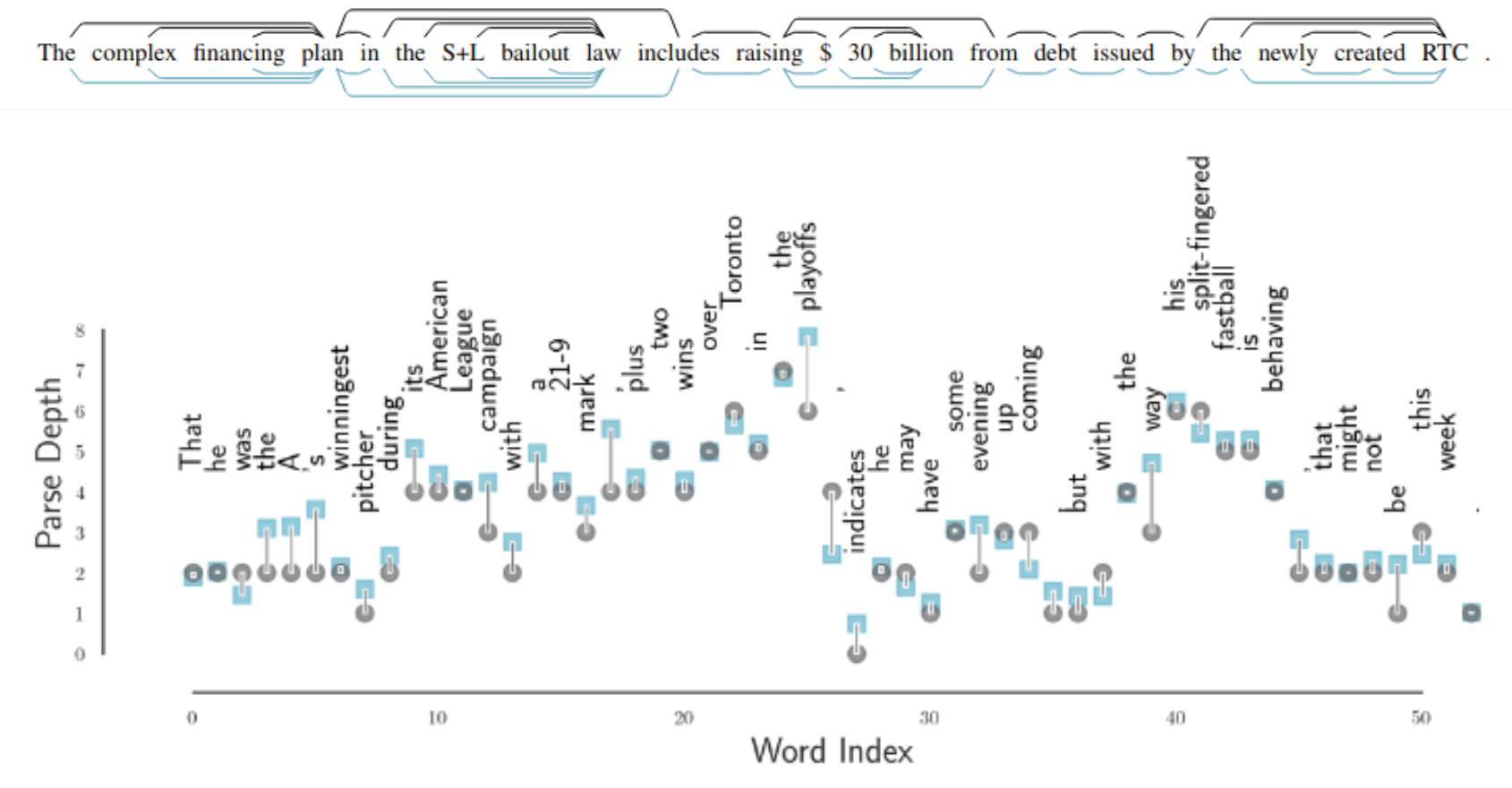
Representation Questions

Syntactic tree decodability



Representation Questions

Syntactic tree decodability



Mechanistic Interpretability

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Mechanistic interpretability identifies the specific **computational circuits** that implement a behavior at the algorithmic level.

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Interpretation

Mechanistic interpretability identifies the specific **computational circuits** that implement a behavior at the algorithmic level.

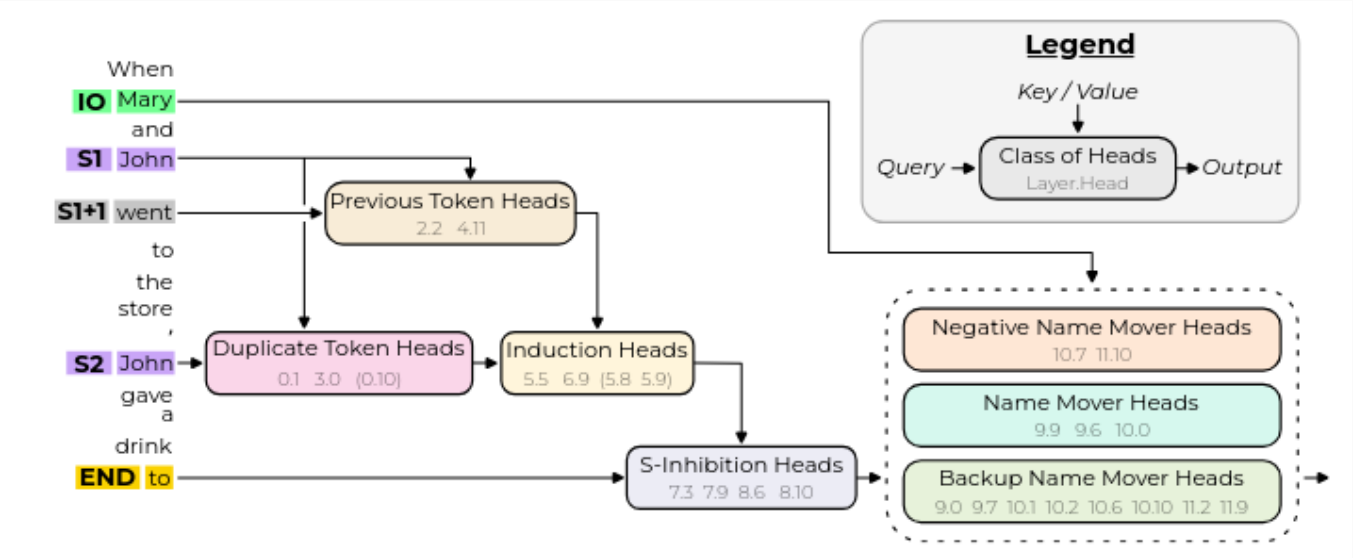
Attention circuits

INTERPRETABILITY IN THE WILD: A CIRCUIT FOR INDIRECT OBJECT IDENTIFICATION IN GPT-2 SMALL

Kevin Wang[1], **Alexandre Variengien**[1], **Arthur Conmy**[1], **Buck Shlegeris**[1] **& Jacob Steinhardt**[1,2]
[1]Redwood Research
[2]UC Berkeley
kevin@rdwrs.com, alexandre@rdwrs.com,
arthur@rdwrs.com, buck@rdwrs.com, jsteinhardt@berkeley.edu

Mechanistic Interpretability

Attention circuits



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- Norm augmentation
- Linguistic annotation
- Stimulus generation
- Experiment scripting

Interpretation

LLMs can function as research tools without requiring direct cognitive interpretation.

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LLMs as Research Tools

Norm augmentation

Behavior Research Methods (2024) 56:6082–6100
https://doi.org/10.3758/s13428-024-02337-z

ORIGINAL MANUSCRIPT

Check for updates

Can large language models help augment English psycholinguistic datasets?

Sean Trott[1]

Accepted: 5 January 2024 / Published online: 23 January 2024
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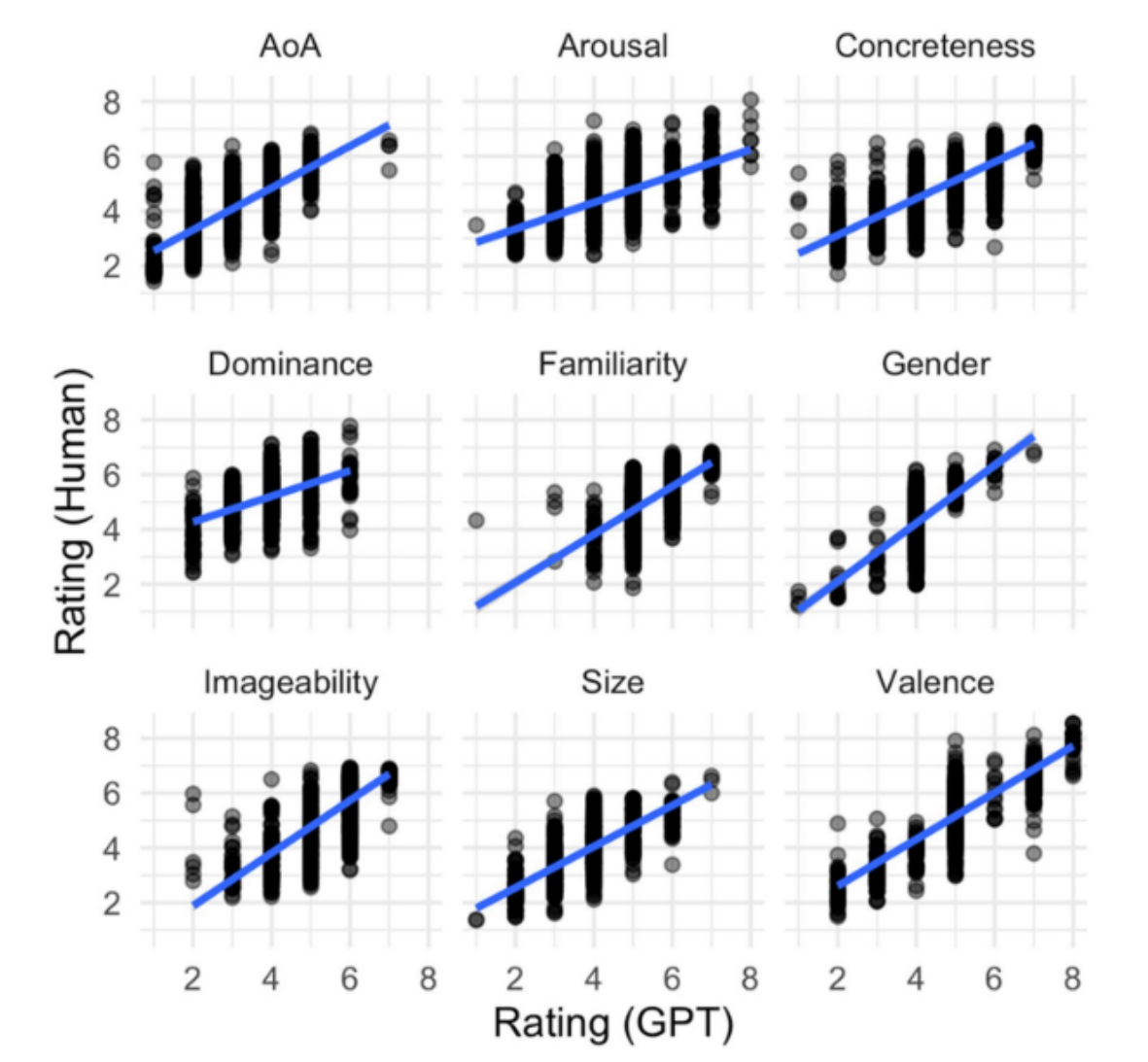
LLMs as Research Tools

Norm augmentation



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